

Let $d \in \mathbb{N}$ be not a perfect square, and let $p, q \in \mathbb{Z}$ satisfy

$$p^2 - dq^2 = 1$$

Suppose that for $n \in \mathbb{N}$,

$$a_n + b_n \sqrt{d} = (p + q\sqrt{d})^n$$

where $a_n, b_n \in \mathbb{Z}$.

Prove by mathematical induction that $a_n^2 - db_n^2 = 1 \quad \forall n \in \mathbb{N}$

Pf. Base Case $n=1$

$$a_1 + b_1 \sqrt{d} = p + q\sqrt{d} \text{ . so } a_1 = p, b_1 = q$$

Hence, $a_1^2 - db_1^2 = p^2 - dq^2 = 1 \Rightarrow P(1)$ holds.

Now assume that for some $k \geq 1$,

$$a_k^2 - db_k^2 = 1$$

$$\text{Since } a_k + b_k \sqrt{d} = (p + q\sqrt{d})^k$$

$$\text{We have } a_{k+1} + b_{k+1} \sqrt{d} = (p + q\sqrt{d})^{k+1} = (p + q\sqrt{d})^k (p + q\sqrt{d})$$

$$\text{Thus, } a_{k+1} + b_{k+1} \sqrt{d} = (a_k + b_k \sqrt{d})(p + q\sqrt{d}).$$

$$= (pa_k + dqb_k) + (qa_k + pb_k)\sqrt{d}$$

$$\text{Therefore, } a_{k+1} = pa_k + dqb_k \quad \& \quad b_{k+1} = qa_k + pb_k$$

$$\text{Now compute: } a_{k+1}^2 - db_{k+1}^2 = (pa_k + dqb_k)^2 - d(qa_k + pb_k)^2$$

$$= p^2 a_k^2 + 2pa_k dqb_k + d^2 q^2 b_k^2 - d(q^2 a_k^2 + 2dpq a_k b_k + d p^2 b_k^2)$$

$$= (p^2 - dq^2) a_k^2 - d(p^2 - dq^2) b_k^2$$

$$= (p^2 - dq^2) (\cancel{a_k^2 - db_k^2}) (a_k^2 - db_k^2)$$

Since $p^2 - dq^2 = 1$ & by the induction hypothesis, $a_k^2 - db_k^2 = 1$

$$\text{We obtain } a_{k+1}^2 - db_{k+1}^2 = 1$$

Therefore, by mathematical induction, $a_n^2 - db_n^2 = 1 \quad \forall n \in \mathbb{N}$.

